

Polynomially long consecutive failures of log-concavity in the bulk of tree independence sequences

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Abstract

For each integer $b \geq 64$, we give a fixed family of trees of maximum degree $b + 1$ whose independence sequences contain a consecutive run of $\Omega(N^{\log_b(bq)})$ strict log-concavity failures at a limiting relative position strictly below 1. Here $q < 1/2$ is an explicitly specified algebraic number. At maximum degree 65, the run has length $\Omega(N^{4/5})$ and lies below 0.99 times the independence number. Allowing the fixed degree bound to depend on ε yields $\Omega(N^{1-\varepsilon})$ consecutive bulk failures for every $\varepsilon > 0$. Consequently the extremal run-length exponent is 1. The analytic step is a finite-height, stretched-exponential Fourier bound; it gives uniform convergence of an exact lattice interpolation and all its derivatives. This closes the transfer from a limiting mass valley to a consecutive block of finite coefficient inequalities.

Status. This is a proposed proof with all analytic steps supplied below, not an independently refereed result or a formal verification. No claim of bibliographic priority is made. The lower-bound constants and the starting height in this stronger run theorem are not numerically evaluated.

1 Statement

For a tree T , let $i_k(T)$ count its independent sets of size k , and write $\alpha(T)$ for its independence number. A failure is an interior index k such that $i_k(T)^2 < i_{k-1}(T)i_{k+1}(T)$.

Fix an integer $b \geq 64$. Let $T_{b,0}$ be an edge rooted at one endpoint. Construct $T_{b,h+1}$ by attaching a new root to the roots of b disjoint copies of $T_{b,h}$. Put

$$N_h = b^h + \frac{b^{h+1} - 1}{b - 1}, \quad \alpha_h = b^h + \sum_{j=0}^{\lfloor (h-1)/2 \rfloor} b^{h-1-2j}. \quad (1)$$

The sum is empty at $h = 0$. These are respectively the order and independence number of $T_{b,h}$; its maximum degree is at most $b + 1$.

Let δ be the unique solution in $(2^{-b}, 2^{1-b})$ of

$$\delta(2 - \delta)^b = 1, \quad r = 1 - \delta, \quad q = \frac{r}{1 + r}, \quad \rho = bq, \quad \pi = \frac{q}{1 + q}. \quad (2)$$

Define

$$A_b = 1 + \frac{b}{b^2 - 1}, \quad L_b = r + \delta\pi + \frac{\pi}{b - 1}, \quad \beta_b = \frac{L_b}{A_b}, \quad \kappa_b = \log_b \rho. \quad (3)$$

Theorem 1. *For every fixed integer $b \geq 64$, there exist constants $c_b > 0$ and $H_b < \infty$ such that, for every $h \geq H_b$, there is an interval of consecutive interior indices I_h with*

$$|I_h| \geq c_b \rho^h = \Omega_b(N_h^{\kappa_b}), \quad (4)$$

$$i_k(T_{b,h})^2 < i_{k-1}(T_{b,h})i_{k+1}(T_{b,h}) \quad (k \in I_h), \quad (5)$$

$$\sup_{k \in I_h} \left| \frac{k}{\alpha_h} - \beta_b \right| \longrightarrow 0. \quad (6)$$

Moreover

$$0 < \beta_b < \frac{(3b-2)(b+1)}{3(b^2+b-1)} = 1 - \frac{2b-1}{3(b^2+b-1)} < 1. \quad (7)$$

For $b = 64$, the entire run lies below $0.99\alpha_h$ for all sufficiently large h , and has length $\Omega(N_h^{4/5})$.

Corollary 2. For every $\varepsilon > 0$, there are constants $\Delta_\varepsilon < \infty$ and $d_\varepsilon > 0$ and a family of arbitrarily large trees of maximum degree at most Δ_ε with $\Omega_\varepsilon(N^{1-\varepsilon})$ consecutive failures, all at indices below $(1 - d_\varepsilon)\alpha(T)$. The degree bound and relative gap are fixed along each family.

Let $M(N)$ denote the largest length of a consecutive run of failures among trees on at most N vertices.

Corollary 3. The extremal logarithmic exponent is exactly one:

$$\lim_{N \rightarrow \infty} \frac{\log M(N)}{\log N} = 1, \quad \text{equivalently } M(N) = N^{1-o(1)}.$$

This does not assert $M(N) = \Theta(N)$.

Context. Galvin's Question 3.1 asks for log-concavity failure at a fixed relative distance from the last coefficient [1]. Consecutive failures have been studied in [2, 3]. Yu's construction gives $\Omega(N^{1/3})$ consecutive failures with indices $k/\alpha(T) \rightarrow 1$; its discussion explicitly raises a bulk analogue. Fang et al. prove central log-concavity for sufficiently large forests [4]; the location here is outside their guaranteed interval. The present argument strengthens the earlier single-failure construction on the same rooted trees. It neither changes that construction nor contradicts unimodality.

2 Exact recursion and its normalized limit

At $\delta = 2^{-b}$ the left side of (2) is below 1; at $\delta = 2^{1-b}$ it is at least $2(1 - b2^{-b}) > 1$. Its derivative has the sign of $2 - (b+1)\delta$, which is positive between these endpoints. Thus the specified solution exists uniquely. In particular

$$0 < \delta < b^{-2}, \quad \frac{49}{100} \leq q < \frac{1}{2}, \quad \sqrt{b} < \rho < b. \quad (8)$$

Use the same activity $\lambda = r/\delta = r(1+r)^b$ at every vertex. An independent set has probability proportional to $\lambda^{|I|}$. The occupied/unoccupied root partition-function ratio is $\lambda/(1+\lambda) = r$ for $T_{b,0}$, and the recursion $R_{h+1} = \lambda/(1+R_h)^b$ makes it exactly r at every height. Thus each isolated rooted subtree has root occupation probability q .

Let X_h be the size under this probability law, let $\mu_h^s = \mathbb{E}(X_h \mid \text{root} = s)$, and let $m_h = \mathbb{E}X_h$. Root conditioning gives

$$\mu_{h+1}^0 = b[(1-q)\mu_h^0 + q\mu_h^1], \quad \mu_{h+1}^1 = 1 + b\mu_h^0.$$

Therefore, with $D_h = \mu_h^1 - \mu_h^0$,

$$D_0 = \delta, \quad D_{h+1} = 1 - \rho D_h, \quad D_h = \theta + a_h, \quad \theta = \frac{1}{1+\rho}, \quad a_h = (\delta - \theta)(-\rho)^h. \quad (9)$$

Here $\delta < \theta$: indeed $\delta < b^{-2} < 1/(1+b/2) < \theta$. The signed normalization a_h removes the alternating growing mode exactly. Define the centered conditional variables and a deterministic correction by

$$Y_h^s = \frac{X_h - \mu_h^s}{a_h} \mid (\text{root} = s), \quad z_h = \frac{D_h}{a_h} = 1 + \frac{\theta}{a_h}.$$

If S_i are independent Bernoulli(q) variables, with conditionally independent copies of the appropriate conditional laws, then

$$\begin{aligned} Y_{h+1}^0 &\stackrel{d}{=} -\rho^{-1} \sum_{i=1}^b (Y_{h,i}^{S_i} + (S_i - q)z_h), \\ Y_{h+1}^1 &\stackrel{d}{=} -\rho^{-1} \sum_{i=1}^b Y_{h,i}^0. \end{aligned} \tag{10}$$

Lemma 4. *The pair $(\mathcal{L}(Y_h^0), \mathcal{L}(Y_h^1))$ converges in the maximum of the two 2-Wasserstein distances to a unique centered finite-variance fixed pair $(\mathcal{L}(Y^0), \mathcal{L}(Y^1))$ of (10) with z_h replaced by 1. Both limiting variances v_s are positive and at most $5/(4b)$.*

Proof. On pairs of centered finite-second-moment laws, call the transformation in (10) $\mathcal{T}_z$. Couple two input pairs optimally in each state, use common S_i , and take independent couplings over i . The differences have mean zero, so their sum's second moment is the sum of their second moments. Consequently

$$d(\mathcal{T}_z\nu, \mathcal{T}_z\tilde{\nu}) \leq \frac{\sqrt{b}}{\rho} d(\nu, \tilde{\nu}), \quad d(\mathcal{T}_z\nu, \mathcal{T}_1\nu) \leq \frac{\sqrt{bq(1-q)}}{\rho} |z - 1|.$$

The space is complete (centering is closed under W_2 convergence), and $\ell := \sqrt{b}/\rho < 1$. The contraction theorem gives the fixed pair. Its distance e_h from the height- h pair obeys $e_{h+1} \leq \ell e_h + C\rho^{-h}$, so $e_h \rightarrow 0$ (indeed $e_h = O_b(\ell^h)$).

Put $t = b/\rho^2 = 1/(bq^2) < 1$. Taking variances in the fixed equations gives

$$v_1 = tv_0, \quad v_0 = t[(1-q)v_0 + qv_1 + q(1-q)], \quad v_0 = \frac{tq(1-q)}{1-t(1-q)-qt^2} > 0.$$

Since $q^2 \geq (49/100)^2 > 6/25$, we have $t < 25/(6b)$. For $b \geq 64$,

$$1 - t(1-q) - qt^2 \geq 1 - t - t^2 \geq 1 - \frac{25}{384} - \left(\frac{25}{384}\right)^2 > \frac{5}{6}.$$

Thus $0 < v_1 \leq v_0 \leq (25/(24b))/(5/6) = 5/(4b)$. □

The unconditional centered normalization is

$$Z_h := \frac{X_h - m_h}{a_h} \stackrel{d}{=} Y_h^S + (S - q)z_h \longrightarrow Z := Y^S + (S - q) \tag{11}$$

in W_2 , where S has probability q of being 1.

3 A Fourier bound uniform at finite height

Weak convergence alone does not decide neighboring coefficient signs. We now control the complete fundamental frequency interval before taking any derivative or lattice difference.

Let $\chi_h^s(u) = \mathbb{E}(e^{iuX_h} \mid \text{root} = s)$ and $Q_h(u) = \max_s |\chi_h^s(u)|$. The exact conditional recursion is

$$\chi_{h+1}^0(u) = ((1-q)\chi_h^0(u) + q\chi_h^1(u))^b, \quad \chi_{h+1}^1(u) = e^{iu}(\chi_h^0(u))^b. \tag{12}$$

In particular, for $1 \leq k \leq h$,

$$Q_h(u) \leq Q_k(u)^{b^{h-k}}. \tag{13}$$

Lemma 5. Set $\gamma = \log b / \log \rho > 1$. There are positive constants K, c and an integer h_0 , depending only on b , such that for every $h \geq h_0$ and $|t| \leq \pi|a_h|$,

$$\max_s |\mathbb{E} e^{itY_h^s}| \leq K e^{-c|t|^\gamma}, \quad |\mathbb{E} e^{itZ_h}| \leq K e^{-c|t|^\gamma}. \quad (14)$$

Proof. Let ψ_s be the characteristic functions of Y^s . Positive finite variance gives $|\psi_s(t)|^2 = 1 - v_s t^2 + o(t^2)$ at zero. Choose $\eta > 0$ sufficiently small that both moduli are strictly below 1 throughout $\eta \leq |t| \leq \rho\eta$. Convergence in W_2 gives uniform convergence of characteristic functions on compact intervals, since $|\psi_h^s(t) - \psi_s(t)| \leq |t|W_2(\mathcal{L}(Y_h^s), \mathcal{L}(Y^s))$. Thus, after increasing h_0 if necessary, there is $u_* < 1$ such that

$$Q_k(t/a_k) \leq u_* \quad (k \geq h_0, \eta \leq |t| \leq \rho\eta). \quad (15)$$

Increase h_0 again to ensure $\eta/|a_{h_0}| < \pi$.

First suppose $\eta/|a_h| \leq |u| \leq \eta/|a_{h_0}|$. Choose $k \in [h_0, h]$ with $\eta \leq |a_k u| \leq \rho\eta$; successive values grow by the factor ρ . Writing $t = a_h u$, we have $|t| \leq \rho\eta\rho^{h-k}$. Hence (13)–(15) give

$$Q_h(u) \leq u_*^{b^{h-k}} \leq \exp\left(-\frac{-\log u_*}{(\rho\eta)^\gamma} |t|^\gamma\right).$$

For the remaining physical frequencies $\eta/|a_{h_0}| \leq |u| \leq \pi$, observe that

$$X_1 \mid 0 \stackrel{d}{=} \text{Bin}(b, 2q), \quad X_1 \mid 1 \stackrel{d}{=} 1 + \text{Bin}(b, r).$$

Indeed X_0 is Bernoulli($2q$), while $X_0 \mid 0$ is Bernoulli(r). Since $0 < 2q, r < 1$, both conditional laws have lattice span one, and $Q_1(u) < 1$ for $0 < |u| \leq \pi$. Compactness gives a common bound $v_* < 1$ on the remaining interval. Let $a_0 = \delta - \theta$. Then

$$Q_h(u) \leq v_*^{b^{h-1}} \leq \exp\left(-\frac{-\log v_*}{b(\pi|a_0|)^\gamma} |t|^\gamma\right),$$

because $|t| \leq \pi|a_0|\rho^h$ and $\rho^\gamma = b$. For $|t| < \eta$ a constant prefactor suffices. Centering does not change modulus; nor does the sign of a_h affect these bounds. The unconditional characteristic function is a convex combination of the centered conditional ones with unit-modulus phase factors. This proves both claims. $\square$

4 Smooth densities and exact lattice interpolation

Lemma 6. The laws of Y^0, Y^1, Z have densities f_0, f_1, f that extend to entire functions and are strictly positive everywhere on the real line. If $\phi_h(t) = \mathbb{E} e^{itZ_h}$ and

$$g_h(x) = \frac{1}{2\pi} \int_{-\pi|a_h|}^{\pi|a_h|} e^{-itx} \phi_h(t) dt, \quad (16)$$

then, for every fixed integer $j \geq 0$,

$$\sup_{x \in \mathbb{R}} |g_h^{(j)}(x) - f^{(j)}(x)| \longrightarrow 0. \quad (17)$$

At each lattice node $x_{h,k} = (k - m_h)/a_h$ one has exactly

$$g_h(x_{h,k}) = |a_h| \mathbb{P}(X_h = k). \quad (18)$$

Proof. Lemma 5 passes to the limiting characteristic functions. Since $\gamma > 1$, the bound $Ke^{-c|t|^\gamma}$ remains integrable after multiplication by $|t|^j e^{R|t|}$ for every finite j, R . Fourier inversion therefore gives densities with entire extensions. On the real axis they are nonnegative. Neither conditional density is identically zero, so its real zeros are isolated and it is positive almost everywhere.

The fixed equation for Y^1 expresses its density as the scaled and reflected b -fold convolution of f_0 . A convolution of two integrable densities positive almost everywhere is strictly positive at every point: its integrand is positive almost everywhere. These densities are bounded, so the convolution is finite. Iteration shows $f_1 > 0$ everywhere. The density of $Y^S + (S - q)$ is

$$f(x) = (1 - q)f_0(x + q) + qf_1(x - (1 - q)) > 0.$$

The fixed equation for Y^0 now expresses f_0 as a scaled reflected b -fold convolution of this positive density, so $f_0 > 0$ everywhere as well. The identities first hold almost everywhere and then everywhere by continuity.

Extend $\phi_h(t)$ by zero outside its fundamental interval in (16). Pointwise convergence and Lemma 5 give, by dominated convergence,

$$\int_{\mathbb{R}} |t|^j |\phi_h(t) \mathbf{1}_{\{|t| \leq \pi|a_h|\}} - \mathbb{E}e^{itZ}| dt \rightarrow 0.$$

Fourier inversion and differentiation under the integral prove (17). The functions g_h are real by conjugate symmetry. Finally, substitute $t = a_h u$ in (16) at $x_{h,k}$. Lattice Fourier inversion over $[-\pi, \pi]$ gives (18); the absolute Jacobian $|a_h|$ handles either sign of a_h . $\square$

5 A valley forces a whole interval of positive log curvature

Lemma 7. *There are a closed interval J of positive length and a constant $\zeta > 0$ such that $(\log f)'' \geq 2\zeta$ throughout J . One can take J inside $[-q - 1/4, 1 - q + 1/4]$.*

Proof. In the mixture $Z = Y^S + (S - q)$, the component means are $-q$ and $1 - q$, their variances are at most $5/(4b) \leq 5/256$, and both weights are at least $49/100$. Consider intervals of radius $1/4$ centered at $-q$, $1/2 - q$, and $1 - q$, denoted J_L, J_M, J_R in order. Chebyshev's inequality gives

$$\min\{\mathbb{P}(Z \in J_L), \mathbb{P}(Z \in J_R)\} \geq \frac{49}{100} \left(1 - \frac{5}{16}\right) = \frac{539}{1600}, \quad \mathbb{P}(Z \in J_M) \leq \frac{5}{16} = \frac{500}{1600}.$$

For the middle interval, each component is at least $1/4$ from its own mean. Endpoints have probability zero.

If a positive continuous density is log-concave on the hull of three ordered intervals of equal length, the middle mass is at least the smaller outer mass. To see this, choose a point in each outer interval whose density is at least that interval's average. Concavity of the logarithm bounds the density at every intervening point below by the smaller chosen density, and integration over the middle interval proves the assertion. The strict inequalities above contradict this property. Since $f > 0$ is smooth, $\log f$ is twice continuously differentiable. If its second derivative were everywhere nonpositive in the hull's interior it would be concave there. It therefore has a point of strictly positive second derivative. Continuity supplies J and ζ . $\square$

By (17) with $j = 0, 1, 2$, positivity of f on J implies that, for all sufficiently large h , $g_h > 0$ and

$$(\log g_h)'' \geq \zeta \quad \text{on } J. \tag{19}$$

Let I_h consist of all integers k whose three nodes $x_{h,k-1}, x_{h,k}, x_{h,k+1}$ belong to J . These integers are consecutive, even when $a_h < 0$, and

$$|I_h| \geq |J||a_h| - 4 \geq c_b \rho^h \tag{20}$$

for all sufficiently large h and a constant $c_b > 0$.

Write $p_{h,k} = \mathbb{P}(X_h = k)$ and $s_h = 1/|a_h|$. The exact integral formula for a centered second difference, together with (18)–(19), yields

$$\begin{aligned} \log \frac{p_{h,k-1}p_{h,k+1}}{p_{h,k}^2} &= \int_{-s_h}^{s_h} (s_h - |u|)(\log g_h)''(x_{h,k} + u) du \\ &\geq \frac{\zeta}{|a_h|^2} > 0 \quad (k \in I_h). \end{aligned} \quad (21)$$

All three probabilities are positive, by (18) and positivity of g_h on J . Thus these are interior indices of the true coefficient support. Finally $p_{h,k} = i_k(T_{b,h})\lambda^k/Z_{T_{b,h}}(\lambda)$, so

$$\frac{i_{k-1}(T_{b,h})i_{k+1}(T_{b,h})}{i_k(T_{b,h})^2} = \frac{p_{h,k-1}p_{h,k+1}}{p_{h,k}^2} \geq \exp\left(\frac{\zeta}{|a_h|^2}\right) > 1. \quad (22)$$

This is a finite-height conclusion valid uniformly at every index in the run. No estimate of the form “pointwise error is smaller than mesh squared” is assumed; the exact interpolation and its C^2 convergence are what justify the signs.

6 Location and extremal consequences

We finish the proof of Theorem 1. At depth j in the b -ary skeleton the occupation probability p_j satisfies $p_0 = q$, $p_{j+1} = q(1 - p_j)$, whence

$$p_j = \pi + (q - \pi)(-q)^j.$$

A terminal skeleton vertex and its extra leaf together have expectation $r + \delta p_h$. Summing over levels,

$$\frac{m_h}{b^h} = r + \delta p_h + \sum_{j=0}^{h-1} b^{j-h} p_j \longrightarrow r + \delta \pi + \frac{\pi}{b-1} = L_b.$$

Formula (1) follows alternatively from conditional maxima $u_0 = v_0 = 1$, $u_{h+1} = b \max(u_h, v_h)$, $v_{h+1} = 1 + bu_h$; induction gives the displayed alternating-level sum. Hence $\alpha_h/b^h \rightarrow A_b$. Since J is bounded and $|a_h|/b^h = |a_0|q^h \rightarrow 0$, the nodes defining I_h satisfy

$$\sup_{k \in I_h} |k - m_h| = O_b(\rho^h), \quad \sup_{k \in I_h} |k/\alpha_h - \beta_b| \rightarrow 0.$$

Since $r + \delta \pi < 1$ and $\pi < 1/3$, we have

$$\beta_b < \frac{1 + 1/[3(b-1)]}{1 + b/(b^2-1)} = \frac{(3b-2)(b+1)}{3(b^2+b-1)} < 1.$$

For $b = 64$ this upper bound is $12350/12477 < 99/100$. Also $\rho \geq 64(49/100) > 30$ and $30^5 > 64^4$, so $\kappa_{64} > 4/5$. This proves every assertion of the theorem.

As $b \rightarrow \infty$, $q \rightarrow 1/2$ and $\kappa_b = 1 + \log q / \log b \rightarrow 1$. Given $\varepsilon > 0$, fix an integer $b \geq 64$ with $\kappa_b > 1 - \varepsilon$ and use $d_\varepsilon = (1 - \beta_b)/2$ and $\Delta_\varepsilon = b + 1$. This proves Corollary 2. Here b is fixed before $h \rightarrow \infty$; no estimate uniform in b is being asserted.

For Corollary 3, fix the same b . The orders satisfy $N_{h+1} = bN_h + 1$. For every sufficiently large budget N , choose the largest h with $N_h \leq N$. Then $N_h > (N-1)/b$, and therefore

$$M(N) \geq c'_b N_h^{\kappa_b} \geq c''_b N^{\kappa_b}.$$

It follows that $\liminf \log M(N)/\log N \geq 1 - \varepsilon$ for every $\varepsilon > 0$. The trivial upper bound $M(N) \leq N$ completes the proof. $\square$

What is and is not quantified. The tree family, activity, exponent and limiting bulk location are explicit. The argument establishes positive run constants and a finite starting height for each fixed b , and proves the required uniform passage to finite coefficients. It does not evaluate the curvature interval J , its lower curvature bound, or the resulting H_b . In particular this paper does not assert that a polynomially long run already occurs at the height 8 used in the earlier single-failure certificate. The inequalities here may be very weak in relative size: the guaranteed logarithmic ratio in (21) is of order ρ^{-2h} .

References

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